

Homework 1. Distinct Numbers and Histograms

How to turn in:

- Write two programs and submit them on the course website
- The names are in the assignment; incorrect names will be graded as 0
- Do NOT use any libraries other than for Scanner and File IO (e.g., do not use ArrayList)

Part A. Distinct Number Count (10 points)

This program should:

1. Read an input file name from a user (note: the input file contains several integer numbers)
2. Display the minimum number among the input values
3. Display a list of distinct values in the input and the number of occurrences of each distinct value
4. Be submitted exactly as HW01a_Distinct.java (**misnamed files will receive a zero**)

For this assignment, you can assume that the number of input values in a file can't be more than 30, and each value will be from -99 to 99.

Consider the content of a sample input file hw01a-1.txt. The first number (5) in the file indicates that the file contains 5 values: (2, 1, 2, -3, 2).

```
5
2
1
2
-3
2
```

Here is the corresponding sample run of the program (user input shown in **bold**). **Carefully notice the output format, as your program must display the result exactly as in the sample run.**

```
Enter input file name: hw01a-1.txt
```

```
Min Number: -3
```

Number	Count
2	3
1	1
-3	1

Consider the content of another sample input file hw01a-2.txt.

```
8
-5
1
5
```

3
10
2
3
5

Here is the corresponding sample run of the program (user input shown in bold):

Enter input file name: **hw01a-2.txt**

Min Number: -5

Number	Count
-5	1
1	1
5	2
3	2
10	1
2	1

Output Format (General Formatting, Applicable to All Parts):

- **Case Sensitivity:** The output must use the exact uppercase, and lowercase letters, as shown in the sample runs.
- **Spacing and Line Breaks:** Ensure the output has the spacing and line breaks correctly, as shown in the sample runs.

Output Format (Part A Table):

- There should be exactly two spaces between the "Number" and "Count" titles.
- Align the numbers precisely under the "Number" and "Count" titles. [Depending on the values, five to seven spaces should be between the numbers in each row. For example, seven spaces after the 1 and five spaces after -99.]

Part B. Histograms (5 points)

This program should:

1. Read an input file name from a user
2. Load the data from the file into an array
3. Draw one row in a horizontal histogram for each of the numbers in the array
4. Be submitted exactly as HW01b_Histograms.java

The file will contain a count of values, followed by that many integers. For example, let's assume that your input file has the following numbers.

5
5

2
1
3
7

A sample run of your program should look exactly as below (user input in **bold**):

Enter input file name: **hw01b-1.txt**

```
===== Horizontal Histogram =====  
5: * * * * *  
2: * *  
1: *  
3: * * *  
7: * * * * * * *
```

Assume that there is another sample input file that includes the following numbers.

4
1
4
5
2

This is the sample run with the file.

Enter input file name: **hw01b-2.txt**

```
===== Horizontal Histogram =====  
1: *  
4: * * * *  
5: * * * * *  
2: * *
```

Output Format (Horizontal Histogram):

- There is a space after each `\*`

Part C. Vertical Histogram (10 points)

Add a new method to the Java file for Part B and have it print a vertical histogram. Note that you will need to know the largest number in the array and print from the top down. You will print spaces instead of asterisks for numbers that don't go up that high.

This program should:

1. Read an input file name from a user
2. Load the data from the file into an array
3. Draw one row in a horizontal histogram for each of the numbers in the array
4. Add this code to HW01b_Histograms.java

Then, modify the main to call the vertical histogram method after calling the horizontal histogram method. Here's one sample run:

Enter a filename: **hw01c-1.txt**

==== Horizontal Histogram ====

```
5: * * * * *
2: * *
1: *
3: * * *
7: * * * * * * *
```

==== Vertical Histogram ====

```
      *
      *
*      *
*      *
*      * *
* *    * *
* * *  * *
-----
5 2 1 3 7
```

Here's another sample run:

Enter a filename: **hw01c-2.txt**

==== Horizontal Histogram ====

```
1: *
4: * * * *
5: * * * * *
2: * *
```

==== Vertical Histogram ====

```
      *
      * *
      * *
      * * *
* * * *
-----
1 4 5 2
```

Output Format (Vertical Histogram):

- There is a space after each `\*` and each number.

Reminder: Include the header comment as described below [Note: you need to include this for every homework - you won't be reminded in the later homework, and you will be responsible for remembering to add the header comment]. If you miss parts, you will get a penalty.

```
/*
 * Title: HW01b_Histograms.java
 * Abstract: This program displays a message Hello World on the screen.
 * Author: Dr. Islam
 * Email: sislam@csumb.edu
 * Estimate: 4 hours
 * Date: 5/5/1955
 */

public class HW01b_Histograms {
    public static void main(String [] args) {
        // your code here
    }
}
```